Curriculum and Instruction

UNIVERSITY OF MINNESOTA
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M. Ed. / Initial Licensure in Science Education: Grades 9-12 Chemistry
Prerequisite Course Evaluation

Candidates are required to have an undergraduate degree from an accredited institution with a major in Chemistry **or** closely related field, or meet the prerequisites below, and with appropriate liberal arts/general education coursework. Courses in each of the content areas are required to meet Standard 1 of MNSEPT: "*The learner understands and can create learning experiences that make these aspects of content matter meaningful for students*" (*https://www.revisor.mn.gov/rules/?id=8710.2000*).

Candidate Name:	**UMN ID#:**	**Email:**	**Date:**
Degree Granting Institution:	**Major(s):**	**Minor(s):**	**Prepared by:**

Chemistry Major Status	Major	Degree Granting Institution/ Degree Year	Chemistry Degree Requirement Check ✔
• An undergraduate chemistry degree from an accredited institution in the U.S. or an equivalent 4 year chemistry undergraduate degree from an accredited international institution.			

CANDIDATES WITHOUT A CHEMISTRY MAJOR MUST MEET COURES FOR AN EQUIVALENT OF AN UNDERGRADUATE MAJOR. Per U of MN program requirements, 39 credits are required for a Chemistry Science degree (including General Psychology). Adjustment can be made for Non U of M degree programs with varying credit requirements.

Content Area	**Requirement**	**U of M Course Options**	**Course/Term/Year**	**Credits /Grade**	**Institution If Other Than Degree Granting**
Psychology	General Psychology	PSY 1001			
Chemistry Principles (two courses, with labs)	Chemistry I (4)	CHEM 1061 & 1065 Or CHEM 1081 & 1065 Or CHEM 1071H & 1075H			
	Chemistry II (4)	CHEM 1062 & 1066 Or CHEM 1082 & 1086 Or CHEM 1072H & 1076H			

Analytical Chemistry (with lab)	Introductory Analytical Chemistry (3)	CHEM 2101			
	Analytical Chemistry Lab (2)	CHEM 2111			
Organic Chemistry (two courses with lab)	Organic Chemistry I (3)	CHEM 2301 or 2331H Or *CHEM 2081 & 2085			
	Organic Chemistry II (3)	CHEM 2302 or 2332H Or *CHEM 2081 & 2085			
	Organic Chemistry Lab (4)	CHEM 2311 or 2312H ** *see below*			
Physical Chemistry (one course)	Physical Chemistry (3)	CHEM 4501 or 4502			
Research in Chemistry	Directed Research (2)	CHEM 4094			
Chemistry Electives	Candidates must have the minimum of 35 credits in Chemistry				

* NOTE: The Organic Chemistry option listed that includes CHEM 2081 & 2085 (one course with lab) is the equivalent of CHEM 2301, 2302 AND 2311 (two courses with lab). These were specifically designed by CSE for CBS students. ** NOTE: If applicants complete CHEM 2081 & 2085 they are NOT required to complete CHEM 2311.

Additional Licensure in Grades 5-8 General Science (Optional)
Prerequisite Course Evaluation

Content Area	Requirement	U of M Course Options	Course/Term/Year	Credits/Grade	Institution
Life Science two courses (with labs)	Biology I	BIOL 1001 or 1009 or 1951/1961			
	Biology II	BIOL 2003/3004 or 1055 or GCD 3022 or 3033			
Physics sequence two courses (with labs)	Physics I	PHYS 1101 or 1221 or 1301W			
	Physics II	PHYS 1102 or 1222 or 1302W			
Chemistry sequence two courses (with labs)	General Chemistry I	CHEM 1061 & 1065 or 1081 & 1065 or 1071H & 1075H			
	General Chemistry II	CHEM 1062 & 1066 or 1082 & 1086 or 1072H & 1076H			

Earth science two courses (with labs)	Physical Geology	ESCI 1001			
	Astronomy	AST 1001			